

A Technical Primer for Whitespace 3

The C – V Reconciliation: Parameters, Surviving Hypotheses, and Core Claims

(with the constraints imposed by Whitespace 2)

Originality Program — Whitespace 3 Theoretical Reconciliation

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1 Purpose and how to read this primer

Whitespace 3 (WS3) is a theoretical paper arguing that two empirically robust but apparently contradictory traditions are reconciled once they are recognized as measuring *distinct* outputs of the same dynamics:

- the **Henrich tradition** (Henrich 2004; Powell–Shennan–Thomas 2009; Derex et al. 2013): larger, better-connected populations accumulate and preserve more *cumulative complexity*; small isolated groups lose it;
- the **Wu–Wang–Evans (WWE) tradition** (Wu–Wang–Evans 2019; Lin–Frey–Wu 2023): smaller teams produce more *per-capita disruptive novelty*; larger teams consolidate.

The reconciliation is that the first tradition measures a quantity we call $C(t)$ (cumulative complexity) and the second measures a quantity we call $V(t)$ (per-capita variance generation), and that *C and V need not move together—under the same dynamics they can move in opposite directions in response to the same intervention* (whitespace_3/docs/conceptual.md, §“The core theoretical move”).

This primer does three things, in order:

- (a) **Fixes an unambiguous vocabulary** for the minimal agent model: every state object, every control parameter, and the two output measures $C(t)$, $V(t)$ (§2–§5).
- (b) **Restates, in that vocabulary, which causal hypotheses survive** the Whitespace 2 (WS2) empirical study—the refined preferential-attachment ossification pathway (13-D) and the endogenous-canon pathway (13-H), against the ruled-out “cosmetic diversity” pathway (13-B)—and encodes the WS2 findings as constraints the model must satisfy (§6).
- (c) **States the core claims WS3 must establish**, in qualitative form but with the *shape* (monotone / threshold / saturating / hump) and *scaling* (in N, ρ, H) that must survive the WS2 observations (§7).

Remark 1.1 (Register). WS3’s deliverable is *robust sign-structure*, not a fitted functional form. Per conceptual.md, “the paper’s value is conceptual clarity, not model realism,” and results must “hold under multiple specifications of conformity pressure κ ” and multiple network topologies. Accordingly, this primer specifies each dynamical operator by its *named functional dependencies and required monotonicities*, and treats every explicit closed form as an illustrative instance (collected in §9). A “claim” below is a statement about signs, limits, and scaling laws—not about a particular formula.

Remark 1.2 (What WS2 fixed, and the one thing it did not). WS2 measured, on $\sim 900k$ CS and Physics papers over 1970–2023, three *collective* time series (all rising): demographic diversity, semantic (topical) diversity, and canonical concentration. Crucially, WS2 measured canonical concentration—the empirical realization of the model’s conformity *driver*—but it did **not** measure any *per-capita* novelty rate. In the vocabulary below, WS2 pins down collective quantities and the driver of κ , and leaves V (the per-capita output) as the central open measurement. This asymmetry structures everything: WS2 constrains the model *through κ and the collective outputs*, and the model in turn *predicts V* .

2 Primitives: state objects

Time is discrete, indexed by generations $t \in \{0, 1, 2, \dots\}$.

2.1 Elements, complexity, and prerequisites

Definition 2.1 (Element universe and complexity). Let $\mathcal{E}$ be a countable universe of *elements* (cultural / intellectual units: concepts, techniques, results). Each $e \in \mathcal{E}$ carries a *complexity level* $\ell(e) \in \mathbb{N}_{\geq 1}$.

Definition 2.2 (Prerequisite structure). There is an acyclic *prerequisite* relation $\prec$ on $\mathcal{E}$. For $e \in \mathcal{E}$ write $\pi(e) = \{e' : e' \prec e\}$ for its *direct prerequisites* and $\Pi(e)$ for its *prerequisite closure* (the transitive downward set). We require $\ell(e) = 1 + \max_{e' \in \pi(e)} \ell(e')$ (with $\ell(e) = 1$ when $\pi(e) = \emptyset$), so level equals depth in the prerequisite DAG. Thus a level- k element “requires possession of prerequisites at complexity 1 through $k-1$.”

Remark 2.1 (Two channels of an element: content vs. attachment). Each element e has (i) a *content*—a location in a topic space, formalized by a map $\tau(e)$ into a metric “semantic” space $\mathcal{T}$ —and (ii) an *attachment*—its prerequisite set $\pi(e)$, i.e. *which* prior elements it builds on. This content/attachment split is the model-internal image of WS2’s *topics vs. wiring* distinction and is load-bearing throughout (§6). “Topical diversity” will be a statistic of $\{\tau(e)\}$; “canonical concentration” a statistic of the attachment graph $\{\pi(e)\}$.

2.2 Agents, concept bases, and the population repertoire

Definition 2.3 (Concept base and coherence). Agent $i \in [N]$ holds a *concept base* $B_i(t) \subseteq \mathcal{E}$. A base is *coherent* if it is downward closed: $e \in B_i(t) \Rightarrow \pi(e) \subseteq B_i(t)$ (hence $\Pi(e) \subseteq B_i(t)$)—one cannot hold an element without its foundations. All dynamical operators below are defined to preserve coherence.

Definition 2.4 (Repertoire and loss). The *population repertoire* is $\mathcal{R}(t) = \bigcup_{i \in [N]} B_i(t)$. Element e is *lost* at t if $e \in \mathcal{R}(t-1) \setminus \mathcal{R}(t)$: no agent carries it forward, so (absent re-innovation) it is gone permanently. The *carrier count* is $M(e, t) = |\{i : e \in B_i(t)\}|$.

2.3 Network and subgroups

Definition 2.5 (Interaction network). Agents occupy an undirected graph $G(t) = ([N], E(t))$ whose edges encode who may learn from whom. Write $\mathcal{N}_i(t)$ for the neighborhood of i and $d_i(t) = |\mathcal{N}_i(t)|$ for its degree. The *local redundancy* of e at i is $m_i(e, t) = |\{j \in \mathcal{N}_i(t) : e \in B_j(t)\}|$ (how many of i ’s teachers hold e).

Definition 2.6 (Subgroups and isolation). Agents are partitioned into *groups* $\{\mathcal{G}_1, \dots\}$. For an agent i in group $\mathcal{G}$, its *internal fraction* is the share of its edges that stay inside $\mathcal{G}$. The *isolation* of a group is the mean internal fraction of its members (see ι , §3); $\iota \rightarrow 1$ is a fully insulated group, $\iota \rightarrow 0$ a fully integrated one.

2.4 Emergent structures: the canon and the diversity statistics

These are *derived* from the state (not free parameters); they are exactly the objects WS2 measured or that the output measures reference.

Definition 2.7 (Canonical weight, canon, and concentration). The *canonical weight* of $e \in \mathcal{R}(t)$ is its dependency load—the frequency with which e appears in the prerequisite closures of currently-held elements,

$$w(e, t) = \frac{1}{Z_t} |\{(i, e'') : e'' \in B_i(t), e \in \Pi(e'')\}|, \quad Z_t \text{ a normalizer}, \quad (1)$$

so a high- w element is a widely load-bearing foundation. The *canon* $K_\alpha(t)$ is the top- α fraction of $\mathcal{R}(t)$ by w . *Canonical concentration* is the inequality of the weight distribution,

$$H(t) = \text{Gini}(\{w(e, t)\}_{e \in \mathcal{R}(t)}) \in [0, 1], \quad (2)$$

the model image of WS2’s *reference-canonicity Gini*. (Chu–Evans-style top- N rank stability across years is an equivalent monotone summary.)

Definition 2.8 (Topical breadth and demographic diversity). *Topical breadth* is a diversity functional of the repertoire’s contents, $W(t) = \text{Div}(\{\tau(e) : e \in \mathcal{R}(t)\})$ (e.g. effective number of content clusters, or

embedding-variance)—the model image of WS2’s rising *semantic/topical diversity*. Agents carry *priors* $\theta_i \in \Theta$ (initial concept bases / heuristics); their diversity $\theta = \text{Div}(\{\theta_i\})$ is the model’s *demographic-plurality* input, with realized diversity $D(t)$. In WS3, θ is an *input* dimension, not a core C – V driver (see Constraint 3).

3 Parameters (control knobs)

Table 1 lists every control parameter unambiguously. A parameter is a quantity the modeler sets; it is distinct from *state* (§2) and from *outputs* (§5).

Symbol	Name	Domain	Meaning
N	population size	$\mathbb{N}_{\geq 1}$	number of agents (fixed, or a control we vary/compare across)
ρ	network density	$[0, N-1]$	mean degree (edges per agent); density fraction = $\rho/(N-1)$
f	transmission fidelity	$[0, 1]$	per-teaching-event probability of successful acquisition
ε	base innovation rate	$[0, 1]$	per-agent, per-generation propensity to attempt a novelty
b	vertical share	$[0, 1]$	fraction of attempted innovations that are <i>depth-extending</i> (vs. lateral)
ι	isolation	$[0, 1]$	mean internal-edge fraction of a subgroup (1 = insulated)
k	persistence filter	$\mathbb{N}_{\geq 1}$	generations a novelty must survive to count in V (ephemera filter)
θ	prior heterogeneity	$\mathbb{R}_{\geq 0}$	diversity of agent priors — the <i>demographic-plurality</i> input
$\kappa(\cdot)$	conformity pressure	(a function)	local drag against novelty; <i>the load-bearing object</i> (§4.3)
$g(\cdot)$	suppression map	$[0, \infty) \rightarrow [0, 1]$	how κ throttles the realized innovation rate; $g \downarrow$, $g(0)=1$

Table 1. The control parameters of the WS3 model. κ and g are *functions* (not scalars); their specification is the crux (§4.3, Core Claim 2).

Remark 3.1 (The Henrich axes vs. the WWE axis). N, ρ, f are the *preservation* axes—they govern how reliably deep structure is transmitted, and feed C directly and monotonically. ι is the *insulation* axis (Claim #17’s structure). κ, g are the *conformity* apparatus—the only machinery that can make per-capita novelty *fall* as the population grows, and hence the only machinery that creates any tension between C and V (§4.3).

4 Dynamics: the generational update

Each generation, every agent applies two operators—*transmission* then *innovation*—each defined to preserve coherence.

4.1 Transmission (acquisition of existing elements)

Definition 4.1 (Acquisition). Agent i acquires an element e held by at least one neighbor with probability governed by fidelity f and local redundancy $m_i(e, t)$:

$$a_i(e, t) = \begin{cases} 1 - (1 - f)^{m_i(e, t)}, & \text{if } \pi(e) \subseteq B_i(t+1) \text{ (prerequisites present),} \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Equation (3) is the model of Henrich–Powell transmission: an element with high redundancy (m large) around a learner is almost surely acquired; an element held by a single low-fidelity teacher is often dropped; an element held by *no* neighbor cannot be acquired, and one held by nobody in $\mathcal{R}$ is lost. *Only the monotonicities matter*: a_i increasing in f and in m_i , with $a_i \rightarrow 0$ as $m_i \rightarrow 0$; (3) is one admissible form.

4.2 Innovation (generation of novel elements)

Definition 4.2 (Innovation attempt). With probability $\varepsilon g(\kappa_i(t))$ (the base rate throttled by conformity, §4.3), agent i generates a novel element e' :

- **Lateral** (prob. $1-b$): a new element at an existing level k , given i holds level- k material. Adds *breadth* to the repertoire’s content $\{\tau(\cdot)\}$.
- **Vertical** (prob. b): an extension to level $k+1$, requiring i to already hold a complete level- k prerequisite chain. Adds *depth*.

The new element is created with an *attachment* $\pi(e') \subseteq B_i(t)$ drawn from what i holds. Define its *canon-alignment*

$$\gamma(e', t) = \frac{|\pi(e') \cap K_\alpha(t)|}{|\pi(e')|} \in [0, 1], \quad (4)$$

the share of its antecedents that are canonical. A *structurally deviant* innovation is one with low γ (it builds on non-canonical foundations); a *canon-aligned* innovation has high γ .

4.3 Conformity pressure κ : the crux

$\kappa_i(t) \geq 0$ is the local drag against *deviating*. It enters only through the suppression map g (decreasing, $g(0) = 1$, $g(\infty) = 0$), and—this is the essential modeling commitment—**it acts on the attachment channel**: it suppresses *structurally deviant* (low- γ) innovation while leaving canon-aligned and purely lateral/content novelty comparatively free. Formally the realized rate of low- γ novelty carries the factor $g(\kappa_i)$, whereas content placement $\tau(\cdot)$ is (to first order) ungoverned by κ .

Definition 4.3 (Candidate mechanisms for κ). Following `conceptual.md`, $\kappa_i(t)$ is specified by one of (at least) three mechanisms, each an increasing function of a local “visible agreement”:

$$\text{(VC) visible consensus: } \kappa_i^{\text{VC}}(t) = \phi\left(\frac{\max_e m_i(e, t)}{d_i(t)}\right), \quad (5)$$

$$\text{(AC) attention competition: } \kappa_i^{\text{AC}}(t) = \psi(N, \text{recognition skew}), \quad (6)$$

$$\text{(CD) canon deviation: } \kappa_i^{\text{CD}}(t) = \chi(H(t)) \cdot (1 - \bar{\gamma}_i(t)), \quad (7)$$

with ϕ, ψ, χ increasing. Mechanism (CD) is the one WS2 makes empirically privileged (§6): its driver is the measured canonical concentration $H(t)$.

Assumption 4.1 (Conformity scaling requirement — the hard constraint). Whatever the mechanism, the population-mean conformity pressure must rise with scale:

$$\frac{\partial \bar{\kappa}}{\partial N} > 0, \quad \frac{\partial \bar{\kappa}}{\partial \rho} > 0, \quad \frac{\partial \bar{\kappa}}{\partial H} > 0, \quad (8)$$

and strongly enough that per-capita novelty is *decreasing* in N beyond a threshold (Core Claim 2). This is non-negotiable: with $\kappa \equiv 0$ the model gives per-capita novelty flat or increasing in N , contradicting WVE and dissolving the reconciliation. κ ’s *scaling is the mechanism that creates the C–V tension; everything else preserves or generates without opposition*.

Definition 4.4 (Generational update). $B_i(t+1)$ is the coherent closure of $[\text{retained}(B_i(t))] \cup [\text{acquired via (3)}] \cup [\text{innovated via 4.2}]$. The canonical Henrich boundary case (generational replacement) sets $\text{retained} = \emptyset$, so every element must be *re-acquired* each generation or be lost; a memory variant retains with a per-element decay. Results are required to be robust to this choice.

5 Output measures: $C(t)$ and $V(t)$

5.1 Cumulative complexity $C(t)$ — a stock of depth

Definition 5.1 ($C(t)$). Let $c_i(t) = \max\{\ell(e) : e \in B_i(t)\}$ be agent i 's supportable depth (coherence guarantees its full prerequisite chain is present). Then

$$C(t) = \max_{i \in [N]} c_i(t) \quad (\text{population max}), \quad \bar{C}(t) = \frac{1}{N} \sum_i c_i(t) \quad (\text{mean-agent-max}), \quad (9)$$

with the *reproducible-frontier* variant $C_{\mathcal{R}}(t) = \max\{\ell(e) : e \in \mathcal{R}(t), \Pi(e) \subseteq \mathcal{R}(t)\}$ when depth may be jointly, rather than individually, held.

C is a **stock**, a **depth**, and a **collective** property. It is evaluated at a single time slice; it counts *preserved* structure whether or not anything new was made this generation; and its persistence is protected by *redundancy*—an element's whole chain must remain jointly reproducible, so C is stabilized by large N , dense ρ , and high f , and is stochastically lost below a critical redundancy (the Tasmania mechanism). C is *not* novelty and *not* per-capita.

5.2 Per-capita variance generation $V(t)$ — a flow of persisting deviation

Definition 5.2 ($V(t)$). Let $\Delta(t) = \mathcal{R}(t) \setminus \bigcup_{s < t} \mathcal{R}(s)$ be the elements appearing for the *first time* at t . Apply the persistence filter: $e \in \Delta(t)$ *counts* iff it survives at least k generations, $e \in \mathcal{R}(t')$ for all $t' \in [t, t+k]$; write $\Delta^{(k)}(t)$ for the survivors. Then

$$V(t) = \frac{|\Delta^{(k)}(t)|}{N}. \quad (10)$$

V is a **flow**, it is **per-capita**, and it is filtered for **persistence**. Two gates matter: novelty (never previously instantiated) *and* survival $\geq k$ generations (a one-off deviation nobody adopts is noise, not variance generation—the model's image of “disruption that stuck”). Because V looks k generations ahead, it is a lagged/retrospective measure.

Definition 5.3 (Structural / lateral decomposition of V). Partition the survivors by attachment novelty (via γ , Eq. (4)) into *structurally deviant* (low- γ) and *canon-aligned/lateral* (high- γ or purely content-novel) sets, giving

$$V(t) = V^{\text{struct}}(t) + V^{\text{lat}}(t). \quad (11)$$

By construction (§4.3) conformity κ suppresses V^{struct} , V^{lat} and the collective breadth W are comparatively free of it. This split is the model's handle on WS2's topics-vs-wiring result.

Remark 5.1 (Why C and V are different animals). C is a stock of *depth*; V is a flow of *per-capita persisting deviation*. Different types, on different quantities, evaluated over different horizons (a slice vs. an interval). Nothing forces them to co-move—which is the whole point. Note also that WS2's rising *topical breadth* W is a *collective* statistic (a cousin of C , on the collective side), *not* V^{lat} : W can rise on the strength of N growing even while per-capita lateral novelty V^{lat} is flat or falling.

6 Which hypotheses survive, in this language

6.1 What WS2 fixed

Table 2 places each WS2 measurement on the ledger. The pattern is: *collective* series and the conformity *driver* are pinned; *per-capita* novelty is unmeasured.

WS2 quantity	Symbol	Ledger side	WS2 finding
demographic diversity	$D(t)$	input (θ)	rising
semantic / topical diversity	$W(t)$	collective (C -family, breadth)	rising
canonical concentration	$H(t)$	driver of κ^{CD}	rising (endogenous)
<i>cumulative depth</i>	$C(t)$	collective (C -family, depth)	<i>not measured</i>
<i>per-capita novelty</i>	$V(t), V^{\text{struct}}, V^{\text{lat}}$	per-capita	<i>not measured</i>

Table 2. WS2’s measurements on the C – V ledger. WS2 constrains the model *through* κ ’s driver H and the collective outputs; V is the open seat (Remark 1.2).

6.2 Pathway verdicts, formalized

WS2 (Phases 2.2–2.3) adjudicated the eight Claim-#13 pathways of the program’s deep-research report.

Table 3 restates each verdict in the present vocabulary. The survivors are **13-D (refined)** and **13-H**.

Pathway	Verbal	Formal correlate	WS2 verdict
13-B	demographic diversification is cosmetic; shared channels flatten output	κ suppresses lateral/content novelty so W collapses while D high: $\partial W / \partial t \leq 0$ with $D \uparrow$	ruled out ($W \uparrow$)
13-D	preferential attachment $\Rightarrow$ attention concentration $\Rightarrow$ ossification	$H(t)$ rises via preferential attachment on the weight w ; <i>refined</i> : concentration is in the attachment graph, not in content $\{\tau\}$, so $W \uparrow$ alongside $H \uparrow$	confirmed (refined)
13-H	canons bootstrap themselves; homogenization needs no intending actor	$H(t)$ rises <i>endogenously</i> from the dynamics, no exogenous forcing	confirmed
13-G	visible consensus suppresses <i>expression</i> , preserves <i>belief</i> (latent/manifest gap)	$V^{\text{struct}} \downarrow$ while W, V^{lat} free — mirrors expression-vs-belief	resonant, <i>untested</i>
13-C	institutional selection rewards tradition	reward-driven rise in canon-alignment $\bar{\gamma}$	compatible, not tested
13-F	measured decline is an estimator artifact	apparent divergence is a denominator/index confound	consistent-with (no spurious decline claimed)
13-A, 13-E	recommender loops; linguistic asymmetry	no corresponding object in the scholarly-citation model	not addressed

Table 3. Claim-#13 pathways in the WS3 vocabulary, with WS2 verdicts. Survivors: 13-D (refined) and 13-H—both statements about how the conformity driver H grows.

Remark 6.1 (The survivors are statements about κ , not about C or V). Refined 13-D and 13-H do not directly assert anything about the outputs C, V . They assert how the *driver* H (hence κ^{CD}) grows: endogenously (13-H) and by preferential attachment on the attachment graph (13-D). The model then *derives* the consequences for

C and V . This is exactly why κ is the load-bearing object (§4.3): WS2’s surviving hypotheses are precisely a measurement of κ ’s driver and of the channel (γ , attachment) on which it acts.

6.3 The WS2 findings as constraints on admissible models

Any WS3 instantiation must additionally satisfy the following—these are not free choices but facts WS2 observed.

WS2 Constraint 1 (Endogenous, scale-driven canon (13-H + refined 13-D)). *$H(t)$ rises endogenously with cumulative scale, via preferential attachment on the canonical weight w : $\partial H / \partial(\text{scale}) > 0$, with no exogenous actuator. Consequently mechanism (CD) is privileged: it must be among the specifications exhibited, and it must reproduce rising H from scale alone. Shape/scaling: H increasing and concave/saturating in field size (Chu–Evans: top- N rank-stability rising from ~ 0.25 toward high values as fields grow; WS2: reference-canoncity Gini rising over 1970–2023).*

WS2 Constraint 2 (κ acts on attachment, not content). *Conformity suppresses the structural channel and (to first order) spares the content channel: $g(\kappa)$ multiplies the low- γ (attachment-deviant) innovation rate, while topic placement $\tau(\cdot)$ and lateral breadth are free. Consequently the model must be able to produce $W(t) \uparrow$ simultaneously with $V^{\text{struct}}(t) \downarrow$.*

WS2 Constraint 3 (No collective collapse — orthogonality, not strict trade-off). *The rise of H must not force a collapse of collective breadth: $\partial W / \partial H \not\propto 0$ as a structural law—indeed WS2 found $H \uparrow$ and $W \uparrow$ jointly, and (Phase 2.3) that a subfield’s H does not predict its demographic–semantic gap. Hence canonical concentration and diversity are independent axes, and demographic plurality θ is an orthogonal input, not a driver of H or a suppressor of W . The model’s reconciliation must therefore be the orthogonality form, not a strict C -vs- V (or H -vs- W) trade-off.*

WS2 Constraint 4 (Differential in canon concentration). *Structural suppression is stronger where H is higher: $\partial V^{\text{struct}} / \partial H < 0$, and the effect scales with H . Empirical anchor: WS2 found the citation-geometry convergence strongest in Physics (the most canon-concentrated field) and absent/reversed in CS under topical embeddings—a field-level gradient the model must reproduce.*

7 Core claims WS3 must establish

Each claim is stated qualitatively, with the *shape* and *scaling* that must survive. “Establish” means: derive analytically where tractable, else demonstrate numerically, and—decisively—show invariant across ≥ 2 –3 specifications of κ and multiple network topologies (Core Claim 4). Write starred quantities C^*, V^* for long-run / steady-state values.

Core Claim 1 (Henrich reproduced: C rises and is redundancy-protected). *C^* is non-decreasing in the preservation axes over the relevant regime,*

$$\frac{\partial C^*}{\partial N} \geq 0, \quad \frac{\partial C^*}{\partial \rho} \geq 0, \quad \frac{\partial C^*}{\partial f} \geq 0,$$

and small/insulated populations (low N , low ρ , high t , low f) suffer stochastic C -loss. Shape: increasing and saturating in N, ρ, f ; a threshold collapse below a critical redundancy m^ (variance of C -loss $\uparrow$ as $m \rightarrow m^*$). Status: reproduce a known tradition. Anchor: Henrich 2004; Powell et al. 2009 (Tasmania).*

Core Claim 2 (WWE reproduced: per-capita V falls with scale, via κ). *There is a threshold N^* beyond which*

per-capita V^* decreases in N ,

$$\frac{\partial V^*}{\partial N} < 0 \quad (N > N^*), \quad \text{and this requires} \quad \frac{\partial \bar{\kappa}}{\partial N} > 0 \quad (\text{Assumption 4.1}).$$

Shape: per-capita V is hump-shaped in N (rising for tiny N where more minds help, then falling as consensus dominates); the small-team advantage is the descending branch. Scaling: the decline tracks $\bar{\kappa}(N, \rho, H)$; with $\kappa \equiv 0$, V^* is flat/increasing—the falsifier of a mis-specified conformity term. Status: reproduce a known tradition. Anchor: Wu–Wang–Evans 2019; Lin–Frey–Wu 2023.

Core Claim 3 (The reconciliation: same lever, opposite signs (and a non-trivial Pareto set)). *There exist interventions that move C and V in opposite directions: canonically, density,*

$$\frac{\partial C^*}{\partial \rho} > 0 \quad \text{while} \quad \frac{\partial V^*}{\partial \rho} < 0,$$

(and likewise for N). The trade-off is not strict: there also exist interventions that raise both—prototypically selective isolation, raising ι for a designated innovation subgroup (protecting its V) while holding global N, ρ for preservation (protecting C): $\exists$ intervention with $\partial C^* > 0, \partial V^* > 0$. Shape: the Jacobian of (C^*, V^*) over the parameter space has both opposite-sign columns (the trade-off) and same-sign columns (the Pareto directions); a phase diagram partitions $(N, \rho, \kappa$ -scaling) into C -optimal vs. V -optimal regions. Status: the paper’s central result (weak form: trade-off frequent, not universal). Anchor: conceptual.md §“Results to demonstrate.”

Core Claim 4 (Robustness / shape-invariance). *Claims 1–3 hold under (i) ≥ 2 –3 specifications of κ (VC / AC / CD of Def. 4.3) and of the suppression map g ; and (ii) random, small-world, and scale-free topologies. The target is the sign-structure and scaling, invariant to functional-form detail. Status: the discipline that makes this a qualitative theorem rather than a fitted model.*

Core Claim 5 (WS2-consistency). *Admissible instantiations satisfy Constraints 1–4: (a) the CD mechanism is exhibited and yields endogenous, scale-driven, saturating H ; (b) κ suppresses V^{struct} while sparing content, so the model reproduces $W \uparrow$ with $V^{\text{struct}} \downarrow$; (c) $H \uparrow$ does not force $W \downarrow$ (orthogonality; θ an orthogonal input); (d) suppression is stronger at higher H ($\partial V^{\text{struct}} / \partial H < 0$). Status: the empirical discipline WS2 imposes—these are observed facts, not modeling latitude.*

Core Claim 6 (The open prediction: what a per-capita- V study should find). *WS2 fixed $D \uparrow, W \uparrow, H \uparrow$ and left V unmeasured. Given those measured facts, the model predicts (it is not entailed by WS2’s measured quantities alone, but is entailed by the model given WS2’s measured κ -behavior):*

$$\frac{dV^{\text{struct}}}{dt} < 0, \quad \text{strongest where } H \text{ is largest (Physics),} \quad \frac{dV^{\text{lat}}}{dt} \text{ need not be negative.}$$

Shape/scaling: a per-capita decline in structurally-novel output, graded by H ; per-capita topical novelty comparatively spared. Status: the falsifiable seat WS2 left open—measurable by a per-capita- V extension of WS2 (per-author/team novelty over time, reusing the embeddings and authorship data). Confirmation supports the reconciliation on real data; disconfirmation (V^{struct} flat/up despite $H \uparrow$) falsifies the CD-mechanism reading.

Amendment (2026-07-07) — Core Claim 6 disconfirmed; superseded by the content channel. The per-

capita- V extension was run (WS2 Phase 2.4). V^{struct} does *not* decline: it **ris**es ($dV^{\text{struct}}/dt > 0$; off-canon share $+0.006$ and reference-atypicality median- z -0.64 , both $p=0.0005$, team- and volume-controlled), and the rise is **cross-subfield**—within a subfield combination-novelty is flat, only against the whole field does novelty rise. This falsifies the *CD-mechanism reading* (that κ 's rise *suppresses* per-capita structural novelty), exactly as the *Status* line anticipated. It does **not** break the reconciliation—it selects the *orthogonality* branch. WS3 rung 4d (subfield.py) reproduces the observed sign from a **content/ τ channel**: niches recombining a shared canon, niche count $K = N/m$ growing with scale, which drives $V^{\text{struct}} \uparrow$ as a *fragmentation* effect *orthogonal* to the attachment channel's $H \uparrow$ (consistent with τ being “to first order ungoverned by κ ,” §4.3). Corrected picture: the field **concentrates at the top** ($H \uparrow$, attachment/ κ) and **fragments in the middle** ($V^{\text{struct}} \uparrow$, content/ τ) *at once*. The model's κ -crossover (Core Claim 2) survives as a *theoretical* capability, but it is **not** the empirical driver of the V^{struct} trend—the content channel is. See whitespace_2/docs/phases/phase-2.4-retro.md and whitespace_3/docs/phases/phase-1-rung4d-retro.md.

8 Summary ledger

Quantity	Symbol	Type	WS2 status	Model role
cumulative complexity	$C(t)$	stock, depth, collective	not measured	Henrich output (CC 1)
per-capita variance	$V(t)$	flow, per-capita	not measured	WWE output (CC 2); open (CC 6)
structural / lateral topical breadth	$V^{\text{struct}}, V^{\text{lat}}$ $W(t)$	flow, per-capita collective (C-family)	not measured rising	κ acts on struct (WSC 2) must not collapse (WSC 3)
canonical concentration	$H(t)$	driver of κ^{CD}	rising, endog.	privileged mechanism (WSC 1)
demographic diversity	$D(t) / \theta$	input	rising	orthogonal input (WSC 3)
conformity pressure	$\kappa(\cdot)$	control (function)	driver measured via H	<i>the crux</i> (Ass. 4.1)

Table 4. The full ledger: what each quantity is, whether WS2 pinned it, and its role in the WS3 reconciliation.

The one-sentence summary: *WS3 must show that a single dynamics—with a conformity term κ whose mean rises with scale and acts on the attachment channel—drives the stock C up (Henrich) while pushing the per-capita flow V down (WWE), robustly across κ -specifications, in a way that reproduces WS2's measured rise of H and W and predicts the still-unmeasured decline of V^{struct} .*

Superseded (2026-07-07): the final clause is disconfirmed—the extension (WS2 Phase 2.4) found V^{struct} *rising*, not declining. Read the summary as: the κ -dynamics drives $C \uparrow$ and $H \uparrow$ on the attachment channel, while a *second, orthogonal* content/ τ channel (fragmentation, $K = N/m$; WS3 rung 4d) drives $V^{\text{struct}} \uparrow$ —concentration and fragmentation at once. See the amendment to Core Claim 6.

9 A minimal admissible instantiation

To show the vocabulary is realizable, one concrete choice (an *instance*, not a commitment): scale-free G with mean degree ρ ; acquisition (3); suppression $g(\kappa) = e^{-\kappa}$; mechanism (CD) with $\kappa_i^{\text{CD}} = \lambda H(t) (1 - \bar{\gamma}_i)$ for a coupling $\lambda > 0$; canonical weight w as dependency load (Def. 2.7); persistence filter $k = 2$. Preferential attachment—new elements attach prerequisites with probability $\propto w$ —makes $H(t)$ rise endogenously (Constraint 1), which raises $\bar{\kappa}^{\text{CD}}$, which throttles low- γ novelty (Constraint 2), delivering $V^{\text{struct}} \downarrow$ while lateral

placement keeps $W \uparrow$ (Constraint 3); redundancy from large N , ρ protects C (Core Claim 1). Swapping g , the κ -mechanism, and the topology, and checking the signs of Table 4 persist, is Core Claim 4.